

Elasticity: The Arithmetic of Responsiveness

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 5** — Elasticity and Its Application.

Read the chapter alongside these slides.

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Slide 2

Working through it

Why this chapter. Mankiw's chapter 5 runs the argument in roughly the order these slides do: why a percentage is the right measure, how the price elasticity of demand is computed, what determines its size, how it varies along a demand curve, the elasticity of supply, and the relationship between elasticity and total revenue.

Two differences worth knowing before you read. Mankiw computes elasticities with the midpoint method alone. These slides add a second route, reading the slope straight off

a demand equation and scaling it by P/Q , and Part 2 uses the agreement of the two as a check. Second, Mankiw places tax incidence in his next chapter. Part 7 here does the elasticity half of it — who bears the burden — and leaves the wedge, the revenue raised and the lost trades to the study of government policy.

How to use it. Read the chapter once before the material and once after. The slides move faster and leave out the worked detail; the text is slower and does not tell you which steps carry the weight. Use the companion for the arithmetic the slides compress.

Lecture outline

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1. **Why elasticity** — percentages, because units lie.
2. **Measuring the response of demand** — three ways to one number.
3. **Along the line, and the extremes** — one curve, every elasticity.
4. **What makes demand elastic** — five determinants.
5. **Total revenue** — when a price rise pays and when it backfires.
6. **The elasticity of supply** — the seller's side.
7. **Tax incidence** — who really pays a tax.
8. **Applications** — wheat and the war on drugs.
9. **Income and cross-price elasticity** — two more dials.

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Slide 3

Working through it

How the nine blocks fit together. The order is not arbitrary, and half a minute spent on the shape makes the rest easier to hold.

Blocks 1–3 build the measure itself. Why a percentage and not a slope; how to compute the price elasticity of demand without ambiguity; and how that number behaves as you move along a single demand curve.

Block 4 turns the question round and asks what makes demand responsive in the first place, so that a good can be judged before any data arrive.

Blocks 5–7 are the payoffs, and they are what the measure was built for. Whether a

price rise raises revenue. The same dial applied to sellers. And who actually bears a tax, whatever the statute says.

Blocks 8–9 apply the tools to two markets and add two further elasticities.

One thread runs through all of it. Everything turns on a comparison of two percentage changes, and on which of the two is larger. If you lose the thread, return to that sentence.

1. Part 1: Why Elasticity

The question supply and demand left open

The question supply and demand left open

A demand or supply shift moves price and quantity in a **predictable direction**. The model said nothing about **size**.

- A cafe raises its price 10%. Does revenue rise or fall?
- A cigarette tax: do smokers pay it, or do the companies?
- A record harvest: are farmers richer or poorer?

Every one of these turns on *how much*, not *which way*.

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Slide 5

Working through it

What this frame is for. It is not teaching a technique. It is showing a hole in the model, so that the measure built in Part 2 has something to do.

What the model does and does not give you. A shift in demand or supply tells you the *direction* price and quantity move. It says nothing about how far. Direction alone answers almost none of the questions people actually ask.

Why each of the three questions needs a size. If the cafe raises its price 10%, revenue rises only if quantity falls by *less* than 10%. The cigarette tax settles on whichever side of the market responds less. And the harvest question is the same arithmetic again: farmers earn

more only if the extra quantity outweighs the lower price. Every one of them is a comparison of two magnitudes.

What is needed. One number, comparable across markets, saying how much quantity responds to price.

Slopes cannot compare markets

Slopes cannot compare markets

The slope of coffee demand is **10 thousand cups per dollar**. The slope of apartment demand is **units per dollar of rent**.

Which market is more price-sensitive? **The slopes cannot say — their units differ.**

Worse: measure the same coffee in *gallons* (16 cups each) and the slope drops from 10 to 0.625. **Same buyers, same behavior, different number.**

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Slide 6

Working through it

Where does 0.625 come from? This is the number the slide asserts and never derives, and it is the whole argument of the frame.

The slope says demand loses 10 thousand *cups* for every extra dollar. Nothing about the buyers changes if we choose to record the same coffee in gallons instead. But there are 16 cups in a gallon, so 10 thousand cups is

$$\frac{10}{16} = 0.625 \text{ thousand gallons}$$

The same fall in coffee drinking is now reported as 0.625 rather than 10.

Why that is fatal. A measure of behavior must not change when you change the ruler. Every buyer did exactly the same thing in both accounts, and the number moved by a factor of sixteen. So the slope is measuring the units as much as the behavior, and cannot

be compared across markets that use different units — which is every pair of markets.

What survives. A percentage. If quantity falls by 20%, it falls by 20% whether you counted in cups, gallons, or barrels, because the unit appears in the numerator and the denominator and cancels. That is the whole motivation for the definition on the next frame.

Percentages are unit-free

Percentages are unit-free

A **10% change** is a 10% change — in cups, gallons, apartments, dollars, or euros.

So measure responsiveness as a **ratio of percentage changes**:

$$\frac{\% \Delta \text{quantity}}{\% \Delta \text{price}}$$

Units cancel. Markets, goods, countries, decades — all become comparable.

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Slide 7

Working through it

The problem this solves. The slope changed when the units changed — 10 in cups, 0.625 in gallons — although no buyer behaved differently. A measure with that property cannot compare two markets.

Why a percentage survives a change of ruler. A percentage change is a change divided by a level, and both are measured in the same units, so the unit cancels:

$$\frac{-10 \text{ thousand cups}}{30 \text{ thousand cups}} = -33.3\% \quad \frac{-0.625 \text{ thousand gallons}}{1.875 \text{ thousand gallons}} = -33.3\%$$

The same fall in coffee drinking, the same number, either way of counting.

Building the ratio. Do that to both variables. The percentage change in quantity is unit-free, the percentage change in price is unit-free, so their ratio is a pure number carrying no

units at all.

What that buys. Pure numbers can be set beside one another. Coffee against apartments, one country against another, one decade against the next — all comparable, which no slope ever is.

The economist's standard dial

The economist's standard dial

Elasticity. A unit-free measure of how much one variable responds to a change in another, expressed as the ratio of their percentage changes.

The family of dials:

- price elasticity of **demand** — the workhorse;
- price elasticity of **supply**;
- **income** and **cross-price** elasticities, at the end.

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Slide 8

Working through it

What the definition says. An elasticity is a ratio of two percentage changes and nothing more. Something moves, something else responds, and you divide one percentage by the other.

Why “dial” is the right picture. The numerator is always the percentage change in a quantity. What changes from one elasticity to the next is only what sits in the *denominator* — which variable you chose to turn.

The four dials built here. Turn the good's own price and you have the price elasticity of demand, which does most of the work. Turn the price facing sellers and you have the price elasticity of supply. Turn the buyer's income and you have the income elasticity. Turn the price of some *other* good and you have the cross-price elasticity.

Why that is worth knowing in advance. The machinery is learned once, on the first of them.

The other three then cost almost nothing.

2. Part 2: Measuring the Response of Demand

The price elasticity of demand

The price elasticity of demand

Price elasticity of demand (E_d). The percentage change in quantity demanded divided by the percentage change in price:

$$E_d = \frac{\% \Delta Q^d}{\% \Delta P}$$

The law of demand makes the two changes **opposite in sign**, so E_d is negative. By convention we report the absolute value $|E_d|$: **bigger number, more responsive buyers**.

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Slide 12

Working through it

Reading the definition aloud. The price elasticity of demand is the percentage change in quantity demanded, divided by the percentage change in the good's own price. Quantity on top, price underneath. Reversing them answers a different question.

Where the minus sign comes from. Demand curves slope downward, so a price rise brings a quantity fall. One percentage change is positive and the other negative, so the ratio comes out negative every time.

Why the sign is dropped. A sign that never varies carries no information. By convention the number is reported as an absolute value, $|E_d|$, which keeps the reading simple: the bigger the number, the more responsive the buyers. Nothing is lost, because the sign was known in advance.

Where the sign is kept instead. On the income and cross-price elasticities in Part 9 the sign genuinely varies, and there it does the classifying. That contrast is the reason for stating

the convention explicitly here.

How to read the number

How to read the number

$ E_d $	A 1% price rise cuts quantity demanded by...
1.8	1.8% — quantity moves <i>more</i> than price
1.0	exactly 1%
0.4	only 0.4% — quantity barely moves

Elasticity is a pure number. No cups, no dollars — so any two markets can be compared.

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Slide 13

Working through it

What the table is doing. No new theory — it is the definition rearranged. From

$$|E_d| = \frac{\% \Delta Q^d}{\% \Delta P} \quad \implies \quad \% \Delta Q^d = |E_d| \times \% \Delta P$$

set the price change to 1% and the quantity change equals the elasticity itself. That is why each row reads straight across.

Row by row. At $|E_d| = 1.8$, a 1% price rise costs 1.8% of quantity: quantity moves more than price. At 1.0 the two move together. At 0.4 the same price rise costs only four tenths of a percent, and buyers barely notice.

The habit to build. Say the sentence every time an elasticity appears: *a one percent price rise costs me this many percent of my quantity.*

What the number is not. It is not cups and not dollars. It is a pure number, which is precisely why it can travel from a coffee market to a housing market without adjustment.

The vocabulary

The vocabulary		
Condition	Term	Buyers...
$ E_d > 1$	Elastic	respond more than proportionately
$ E_d = 1$	Unit elastic	respond exactly in proportion
$ E_d < 1$	Inelastic	respond less than proportionately
$ E_d = 0$	Perfectly inelastic	do not respond at all
$ E_d \rightarrow \infty$	Perfectly elastic	flee any price rise entirely

The boundary is **one**: quantity moving faster or slower than price.

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Slide 14

Working through it

Five words, all pegged to one number. Every term here is defined by where the elasticity sits relative to 1.

Why one is the dividing line. At $|E_d| = 1$ quantity moves by exactly the same percentage as price. Above one, quantity moves proportionately more — **elastic**. Below one, proportionately less — **inelastic**. The boundary is not a convention: it is the point where the two percentage changes are equal.

The two ends of the dial. At $|E_d| = 0$ quantity does not respond at all, however far price moves — **perfectly inelastic**. As $|E_d|$ grows without limit, the faintest price rise drives quantity to nothing — **perfectly elastic**. Both are drawn in Part 3.

A question worth holding on to. Why should *one*, of all numbers, deserve to be the boundary of an entire vocabulary? It gets an exact answer in Part 5, where total revenue turns out to peak precisely there.

A problem: the answer depends on the direction**A problem: the answer depends on the direction**

Two points on the coffee curve $Q^d = 70 - 10P$: $A = (\$4, 30)$ and $B = (\$5, 20)$.

Walking...	% ΔQ	% ΔP	ratio
$A \rightarrow B$	$-10/30 = -33.3\%$	$+1/4 = +25\%$	1.33
$B \rightarrow A$	$+10/20 = +50\%$	$-1/5 = -20\%$	2.50

Same two points, two elasticities. The base of each percentage changed.

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Slide 15

Working through it

Where the two points come from. Both sit on the coffee demand curve $Q^d = 70 - 10P$. At $P = 4$: $Q = 70 - 40 = 30$. At $P = 5$: $Q = 70 - 50 = 20$.

Row 1, walking $A \rightarrow B$. A percentage change is (change)/(starting value). Quantity falls by 10, and we *started* at 30, so $-10/30 = -33.3\%$. Price rises by 1 from a start of 4, so $+1/4 = +25\%$. The ratio is $33.3/25 = 1.33$.

Row 2, walking $B \rightarrow A$. Exactly the same two points, but now we start at B . Quantity rises by 10 from a start of 20, so $+10/20 = +50\%$. Price falls by 1 from a start of 5, so $-1/5 = -20\%$. The ratio is $50/20 = 2.50$.

Why the answers differ. Nothing about the market changed. The *bases* of the percentages changed: 30 and 4 walking one way, 20 and 5 walking the other. Dividing the same change by a different base gives a different percentage.

Why this matters more than it looks. A measurement that depends on which direction you happened to walk is not a measurement at all — two economists with the same data would report different elasticities. The next two frames are the two standard repairs.

Fix 1 — the midpoint method

Fix 1 — the midpoint method

Use the **average** of the two endpoints as the base, in both numerator and denominator:

$$|E_d| = \left| \frac{(Q_2 - Q_1) / \left[\frac{Q_1 + Q_2}{2} \right]}{(P_2 - P_1) / \left[\frac{P_1 + P_2}{2} \right]} \right|$$

For $A = (4, 30)$, $B = (5, 20)$:

$$\frac{10/25}{1/4.5} = \frac{40.0\%}{22.2\%} = \mathbf{1.80} \quad \text{— either direction.}$$

Use it whenever the data are two observed points.

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Slide 16

Working through it

The idea in one line. The problem was that the two directions used different bases. So refuse them the choice: use the *same* base in both directions, and the natural candidate is the average of the two endpoints.

Building the numerator. The quantity change is $|30 - 20| = 10$. The average quantity is $(30 + 20)/2 = 25$. So the percentage change in quantity is $10/25 = 40\%$.

Building the denominator. The price change is $|5 - 4| = 1$. The average price is $(5 + 4)/2 = 4.50$. So the percentage change in price is $1/4.50 = 22.2\%$.

The ratio. $40.0/22.2 = 1.80$.

Why it now works in either direction. Swap the labels A and B and nothing in the calculation changes: the differences flip sign but we take absolute values, and the averages are unaffected by which point you call first. The direction problem is gone by construction, not by luck.

When to reach for it. Use the midpoint method whenever your data are two observed price–quantity pairs and you have no equation. It is the method for real data.

Fix 2 — the slope method, when you have the equation**Fix 2 — the slope method, when you have the equation**

From $Q^d = 70 - 10P$: the coefficient **10** says each \$1 of price moves quantity by 10 (thousand cups). That is the slope — **read it straight off the equation**.

Scale it into percentage terms:

$$|E_d| = \underbrace{10}_{\text{slope}} \times \frac{P}{Q}$$

At the point $(P, Q) = (4.50, 25)$: $|E_d| = 10 \times \frac{4.50}{25} = \mathbf{1.80}$.

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Slide 17

Working through it

Where the 10 comes from. It is already written in the demand equation. In $Q^d = 70 - 10P$, the coefficient on P says that each extra dollar of price costs 10 thousand cups. That is the slope, and no calculation is required to obtain it — you read it off.

Why the slope alone is not the elasticity. The slope is in cups per dollar; elasticity must be a pure number. Multiplying by P/Q is what converts one into the other:

$$\underbrace{10}_{\text{thousand cups per dollar}} \times \underbrace{\frac{P}{Q}}_{\text{dollars per thousand cups}} = \text{a pure number}$$

The units cancel, which is exactly what the previous part demanded.

The worked point. At $(P, Q) = (4.50, 25)$:

$$|E_d| = 10 \times \frac{4.50}{25} = \frac{45}{25} = 1.80$$

What it measures that the midpoint method does not. This gives the elasticity *at a single point*, not across a stretch of the curve. Use it when you hold the equation; use the midpoint method when you hold two observations.

Two routes, one answer

Two routes, one answer

The point (4.50, 25) is the **center of the arc** from A to B .

Midpoint method across the arc $A \rightarrow B$	1.80
Slope method at the arc's center	1.80

Two independent routes agree. If yours ever disagree, one of them is wrong — go looking.

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Slide 18

Working through it

Why the point (4.50, 25) was chosen. It is not arbitrary. It is the exact center of the arc from $A = (4, 30)$ to $B = (5, 20)$: the average price is $(4 + 5)/2 = 4.50$ and the average quantity is $(30 + 20)/2 = 25$.

Those are precisely the two averages the midpoint method uses as its bases. So the slope method evaluated *there* and the midpoint method evaluated *across the arc* are asking the same question about the same place on the curve, by two independent routes.

Why the agreement is worth something. The two calculations share no step. One divides differences by averages; the other reads a coefficient off an equation and scales it. If both routes hit 1.80, it is very unlikely that both went wrong in the same way.

How to use this in practice. When a problem gives you both an equation and two observed points, run both methods. If they agree, you are almost certainly right. If they disagree, one of them contains an arithmetic error and you now know to go and find it — which is worth far more than a single unchecked answer.

3. Part 3: Along the Line, and the Extremes

Plug the numbers in — elasticity along the coffee curve

Plug the numbers in — elasticity along the coffee curve				
$Q^d = 70 - 10P$, slope 10 everywhere. Yet:				
P (\$)	Q (000s)	Substitute $10 \times P/Q$	$ E_d $	Verdict
6	10	$10 \times 6/10$	6.00	elastic
5	20	$10 \times 5/20$	2.50	elastic
4	30	$10 \times 4/30$	1.33	elastic
3.50	35	$10 \times 3.5/35$	1.00	unit elastic
3	40	$10 \times 3/40$	0.75	inelastic
2	50	$10 \times 2/50$	0.40	inelastic
1	60	$10 \times 1/60$	0.17	inelastic

One straight line. Seven different elasticities.

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Slide 22

Working through it

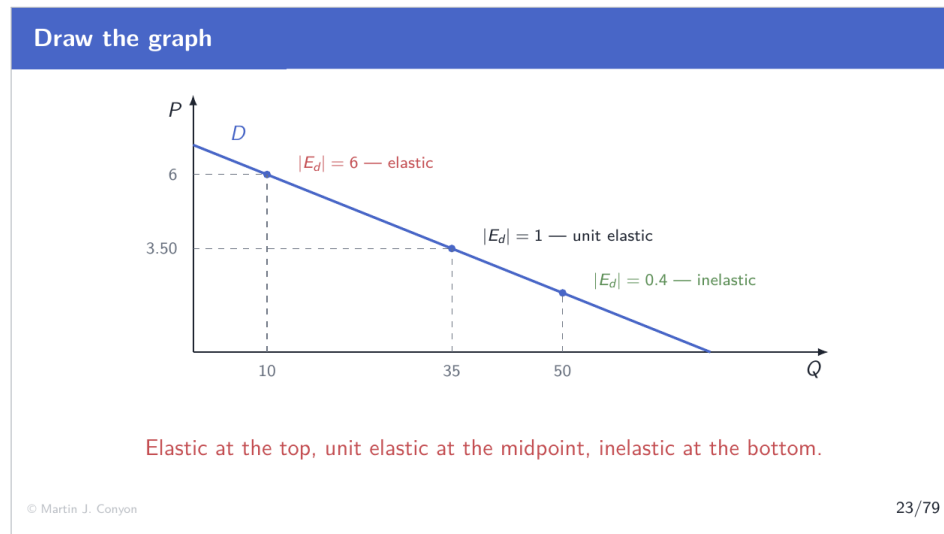
Reading one row. Take the $P = 3$ row and rebuild it. From $Q^d = 70 - 10P$: $Q = 70 - 30 = 40$. Then the slope method: $|E_d| = 10 \times 3/40 = 30/40 = 0.75$. Below one, so inelastic.

The whole column is the same arithmetic. The slope is 10 in every single row — that is what a straight line means. The only thing that changes down the table is the fraction P/Q .

Why P/Q changes so dramatically. As you climb the curve, P rises *and* Q falls, so the ratio is squeezed from both ends at once. At the bottom, $1/60$ is tiny. At the top, $6/10$ is large. Multiplied by 10 that runs from 0.17 to 6.00 — a factor of thirty-five, on a line whose slope never moved.

Where the unit-elastic row comes from. Set the elasticity to one and solve. We need $10 \times P/Q = 1$, that is $Q = 10P$. Substituting the demand equation, $70 - 10P = 10P$, so $70 = 20P$ and $P = 3.50$, giving $Q = 35$. Exactly halfway between the choke price of \$7 and zero — and halfway along the quantity axis too. That is a property of every straight-line demand curve, not a coincidence of these numbers.

Draw the graph



Slide 23

Working through it

What is on the axes. Quantity horizontally, price vertically. This is the ordinary demand curve from the table above, with three of its rows marked.

Where the three marked points come from. They are rows of the table. (10, 6) is the $P = 6$ row, where $|E_d| = 6$. (35, 3.50) is the unit-elastic row. (50, 2) is the $P = 2$ row, where $|E_d| = 0.4$. The dashed guides simply drop each point onto the two axes so you can read its coordinates.

What to look at. The three labels sit on *one straight line*. The line has a single slope; the labels report elasticities differing by a factor of fifteen. Seeing that on one picture is the point of the frame.

Why elastic at the top. High up, quantity is small — only 10 thousand cups. A change of a few thousand cups is therefore a huge *percentage* of quantity, while the same dollar is a small percentage of a high price. Large numerator, small denominator: a big elasticity.

Why inelastic at the bottom. The reverse. Quantity is large, so a change is a small percentage of it; price is low, so a dollar is a large percentage of it. Small numerator, large denominator: a small elasticity.

And in the middle they balance. That balance point is the unit-elastic midpoint, and it is

the same \$3.50 the algebra produced above.

Slope is not elasticity

Slope is not elasticity

Slope — units per dollar. Constant along a straight line.

Elasticity — percentage per percentage. Changes along the same line, because P/Q changes.

“The curve is a straight line, so elasticity is constant” — the classic exam error. You now know why it is wrong.

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Slide 24

Working through it

Two different animals. Slope is a quantity response per dollar of price — 10 thousand cups per dollar on the coffee curve. Elasticity is a percentage per percentage. One carries units, the other does not.

What is constant and what is not. Along a straight line the slope never changes: that is what straight means. Elasticity does change, because

$$|E_d| = |\text{slope}| \times \frac{P}{Q}$$

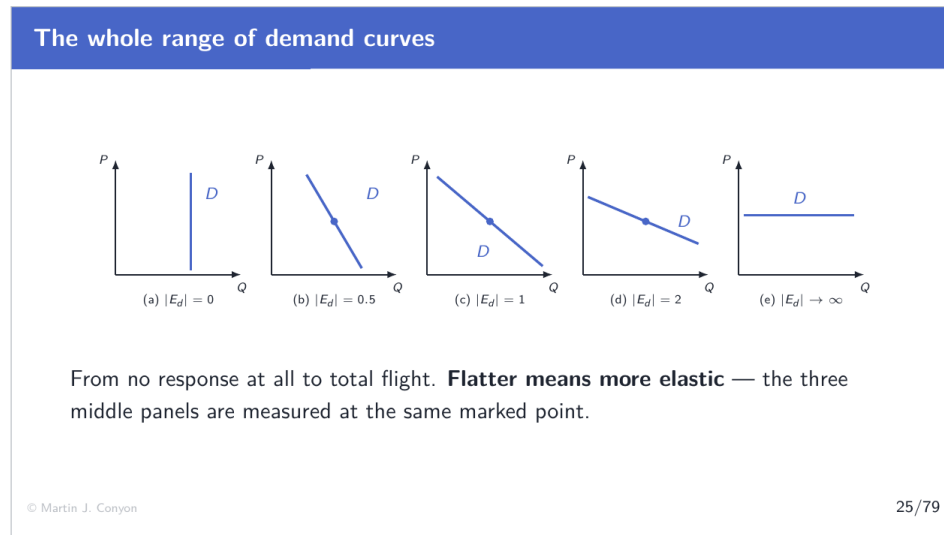
and the fraction P/Q moves as you slide along the line. Climbing the curve, P rises *and* Q falls, so the fraction is squeezed upward from both ends at once.

Why the exam sentence fails. “The curve is a straight line, so elasticity is constant” treats the two as one thing. The coffee table ran from 6.00 down to 0.17 on a single line whose slope was 10 in every row.

The one thing to carry. Constant slope, changing elasticity. They are never the same

statement.

The whole range of demand curves



Slide 25

Working through it

How to read five panels at once. All five have quantity horizontally and price vertically. They are arranged from no responsiveness on the left to total responsiveness on the right.

Panel (a), vertical. Whatever the price, quantity is the same. Buyers do not respond at all, so the percentage change in quantity is zero however large the price change: $|E_d| = 0/\text{something} = 0$.

Panels (b) to (d). As the curve gets flatter, the same vertical price movement produces a larger horizontal quantity movement, so the numerator of the elasticity grows relative to the denominator. All three are drawn through the *same* marked price and quantity, and the number under each panel is the elasticity at that dot: 0.5, 1, and 2. Marking a common point is what makes the comparison mean anything — a straight demand curve does not have a single elasticity, as the last two frames showed, so “this line is inelastic” would be a statement about nothing in particular.

Panel (e), horizontal. Buyers will take any amount at this price and nothing at all above it. The tiniest price rise drives quantity to zero, so the percentage change in quantity is

unbounded: $|E_d| \rightarrow \infty$.

The caveat that keeps this honest. “Flatter means more elastic” is safe when comparing two *different* curves at the same point. It is *not* safe within one curve — you have just watched a single straight line run from 6.00 to 0.17 without changing its steepness at all. The visual rule and the arithmetic rule answer different questions.

The two extremes are real

The two extremes are real

Perfectly inelastic ($|E_d| = 0$, vertical). A transplant patient's anti-rejection medication: the prescribed dose, at almost any price. **Price changes move no quantity at all.**

Perfectly elastic ($|E_d| \rightarrow \infty$, horizontal). One wheat farmer among thousands: charge a cent above the market and sell **nothing**.

The horizontal case is the demand curve facing a single seller in a market with thousands of identical rivals.

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Slide 26

Working through it

Why this frame exists. The two end panels of the figure look like textbook curiosities. They are not. Each describes a real situation, and each has a reason behind it.

Perfectly inelastic, the vertical case. A transplant patient takes the prescribed dose. There is no substitute and no option to take less, so quantity is set by the prescription rather than by the price. The demand curve is vertical and $|E_d| = 0$.

Perfectly elastic, the horizontal case. One wheat farmer among thousands sells a product identical to everyone else's. A cent above the going price and every buyer goes elsewhere, so quantity collapses to nothing; a cent below is money given away. That farmer faces a horizontal demand curve.

Why the second picture is worth keeping. It is not special to wheat. It is the demand curve

facing *any* small seller of a standardized good in a market crowded with identical rivals, which is why it reappears wherever competition is analyzed.

4. Part 4: What Makes Demand Elastic

Substitutes, and how you draw the market

Substitutes, and how you draw the market

1. Close substitutes. Butter's price rises; buyers switch to margarine. Eggs have no stand-in, so egg buyers stay. **More substitutes → more elastic.**

2. Market definition. *Food* has no substitute at all — demand inelastic. *One bakery's sourdough loaf* has a substitute in every other bakery — demand highly elastic. **Narrower market → more elastic.**

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Slide 30

Working through it

The question has changed. Parts 2 and 3 measured elasticity. This part asks what makes it large or small in the first place, so that a good can be judged before any data are seen.

Determinant 1: close substitutes. This one does most of the work. Butter's price rises and buyers slide across to margarine, so butter demand is elastic. Eggs have no close stand-in for most of their uses, so egg buyers stay put and egg demand is inelastic. More escape routes, more elasticity.

Determinant 2: how widely the market is drawn. The same idea from another angle. *Food* as a whole has no substitute — there is no switching away from eating — so its demand is inelastic. *One bakery's sourdough loaf* has a substitute in every other bakery on the street, so its demand is highly elastic. Same bread, different boundary, opposite verdict.

The practical consequence. "Is demand for this elastic?" cannot be answered until you say

demand for what, exactly. Narrow the market and the elasticity rises.

Necessity, luxury, and the size of the bill

Necessity, luxury, and the size of the bill

3. Necessities vs. luxuries. A doctor's visit happens at almost any price; a sailboat does not. **Luxury** → **more elastic** — and which is which lives in the buyer's preferences, not the good.

4. Share of income. Rent doubling rearranges your life; salt doubling goes unnoticed. **Bigger budget share** → **more elastic**.

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Slide 31

Working through it

Determinant 3: necessities against luxuries. A patient sees the doctor at nearly any price, so that demand is inelastic. A sailboat buyer can simply walk away, so that demand is elastic.

The care this determinant needs. Necessity and luxury are facts about the *buyer*, not about the object. The same guitar is a luxury to a weekend player and a necessity to a working musician, and the two elasticities differ accordingly. Never classify a good without asking for whom.

Determinant 4: share of the budget. Doubling the rent rearranges an entire budget, so people respond — they find a roommate, or move. Doubling the price of salt is invisible against everything else, so nobody does anything.

Why the size of the bill matters at all. Responding is costly: it takes searching, switching and thinking. On a large bill the saving repays that trouble; on a small one it does not, so buyers do not bother, and the measured elasticity is low.

Time to adjust

Time to adjust

5. Time horizon. Gasoline jumps today: this month you still drive to work. Over years — a smaller car, a carpool, a move closer to work.

Demand is more elastic in the long run: substitution takes time.

Short-run and long-run elasticity can differ severalfold. **Any policy priced off the short-run number will misfire in the long run.**

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Slide 32

Working through it

Determinant 5: time. The month gasoline jumps, you drive exactly as before, because the car and the commute are already fixed. Over several years you may own something smaller, share the ride, or live closer in.

Same shock, larger response. Nothing about the price change differs between those two accounts. What differs is how much of your life you have had time to rearrange. Substitution is not instantaneous, so the long-run elasticity is the larger of the two.

How much larger. Short-run and long-run estimates for the same good routinely differ several times over. That is wide enough to reverse a conclusion, not a rounding detail.

The warning at the bottom of the slide. A policy built on a short-run elasticity is built on the smaller number. A tax or a price change that looks harmless in its first month can produce a far bigger response once buyers have had time to adjust, so the horizon of the estimate should match the horizon of the policy.

The determinants, measured

The determinants, measured	
Good	Estimated $ E_d $
Eggs	0.1
Healthcare	0.2
Cigarettes	0.4
Rice	0.5
Housing	0.7
Beef	1.6
Restaurant meals	2.3

Necessities at the top, discretionary goods at the bottom. **Treat any single estimate with caution.**

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Slide 33

Working through it

Where these numbers come from. Unlike everything else in these slides, this column is not calculated — it is *estimated*, from real market data, by econometricians. You cannot derive 0.1 for eggs from any equation on these slides. Somebody observed egg prices and egg sales over many periods and fitted the relationship.

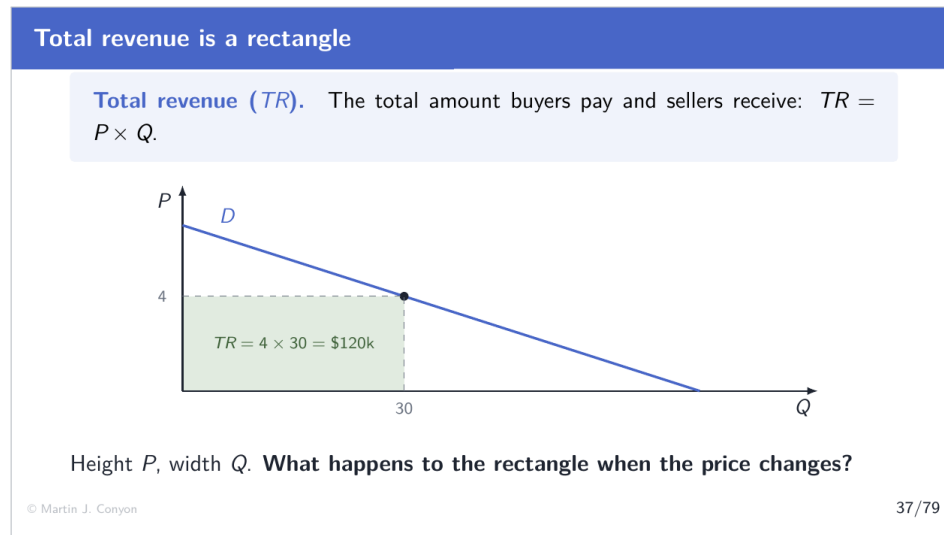
How to read the table anyway. Work down it and check each entry against the five determinants just covered. Eggs at 0.1: no close substitute, small share of income — inelastic, as predicted. Restaurant meals high up the table: an abundance of substitutes, including simply eating at home, and a noticeable share of the budget.

The one worth arguing about. Cigarettes at 0.4. Addiction makes substitution hard in the short run, so demand is inelastic — which is exactly why cigarette taxes raise revenue rather than stopping smoking, and exactly why smokers rather than tobacco companies end up paying them. That rule is derived properly in the tax part, on an invented pizza market; cigarettes are the same arithmetic with different numbers.

The honest caution. Every one of these is a short-run estimate for a particular population in a particular period. Elasticities are not constants of nature like the speed of light. Treat them as measurements, with all that implies.

5. Part 5: Total Revenue

Total revenue is a rectangle



Slide 37

Working through it

What the figure shows. Quantity runs along the horizontal axis and price up the vertical one, as always. The blue line is the coffee demand curve, and it slopes downward for the reason it always does: at a higher price, buyers take less.

Where the shaded rectangle comes from. Pick any point on the demand curve. Its height is the price and its width is the quantity, so its area is price times quantity — which is exactly total revenue. Revenue is not an extra concept bolted onto the diagram; it is an area already sitting inside it.

Where the two numbers come from. The marked point is the equilibrium of the coffee market: \$4 and 30 thousand cups. Multiply them, $4 \times 30 = 120$, and the shaded area is \$120 thousand per week.

The question the figure sets up. Now imagine sliding the marked point up the curve. The rectangle gets *taller*, because the price is higher, and *narrower*, because fewer cups sell. Whether the area grows or shrinks is not obvious from the picture, and that is the whole business of this part.

Two effects fight

Two effects fight

Raise the price and:

- the **price effect** — each unit sells for more — pushes *TR* **up**;
- the **quantity effect** — fewer units sell — pushes *TR* **down**.

	Price rises	Price falls
Elastic ($ E_d > 1$)	<i>TR</i> falls	<i>TR</i> rises
Unit elastic ($ E_d = 1$)	<i>TR</i> unchanged	<i>TR</i> unchanged
Inelastic ($ E_d < 1$)	<i>TR</i> rises	<i>TR</i> falls

Elastic: quantity effect wins. Inelastic: price effect wins.

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Slide 38

Working through it

Why there are two effects and not one. Revenue is $P \times Q$. Raise the price and you change both factors at once, in opposite directions. Each cup still sold earns more — that is the **price effect**, pushing revenue up. But fewer cups sell — the **quantity effect**, pushing revenue down. The product can go either way, and nothing so far tells you which.

Why elasticity settles it. Elasticity is defined as the percentage change in quantity divided by the percentage change in price. So it measures precisely the thing in dispute: whether quantity moves proportionately more or less than price.

If $|E_d| > 1$, quantity moves *more* in percentage terms than price, so the quantity effect is the larger of the two and wins.

If $|E_d| < 1$, quantity moves *less*, so the price effect wins.

If $|E_d| = 1$, they move by exactly the same percentage and cancel.

How to read the table. Take the top-left cell. Demand is elastic, the price rises, the quantity effect wins, and the quantity effect pushes revenue down — so revenue falls. Every other cell is the same three-step reasoning. There is nothing here to memorise: work out which effect wins, then read off which way that effect pushes.

Watch elastic demand punish a price rise**Watch elastic demand punish a price rise**

$|E_d| = 2$. Start: 100 units at \$10 $\Rightarrow TR_1 = \$1,000$.

Raise the price 20% to \$12:

$$\% \Delta Q = 2 \times 20\% = 40\% \text{ fall} \Rightarrow Q = 100 - 40 = 60$$

$$TR_2 = 12 \times 60 = \$720$$

Revenue fell \$280. The 40 lost units overwhelmed the extra \$2 on the 60 that remained.

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Slide 39

Working through it

Line by line.

The starting point is given: 100 units at \$10, so $TR_1 = 10 \times 100 = \$1,000$.

Where does 40% come from? This is the step the slide performs without comment. Elasticity is defined as

$$|E_d| = \frac{\% \Delta Q}{\% \Delta P}$$

We know two of these three quantities: $|E_d| = 2$ and $\% \Delta P = 20\%$. Multiply both sides by $\% \Delta P$ to get the third:

$$\% \Delta Q = |E_d| \times \% \Delta P = 2 \times 20\% = 40\%$$

The elasticity is doing exactly what it was built to do: converting a known price change into the quantity change it causes.

Where does 60 come from? A 40% fall from 100 units removes $0.40 \times 100 = 40$ units, leaving $100 - 40 = 60$.

Where does \$720 come from? New revenue is the new price times the new quantity: $12 \times 60 = 720$.

Reading the result. Revenue fell by \$280. It is worth seeing where that figure hides. The firm *gained* \$2 on each of the 60 units it still sells, which is +\$120. It *lost* 40 units that were earning \$10 each, which is −\$400. Net: $120 - 400 = -\$280$. The two effects are both real and the second is larger.

Watch inelastic demand reward the same move

Watch inelastic demand reward the same move

$|E_d| = 0.5$. Same start: 100 units at \$10 $\Rightarrow TR_1 = \$1,000$.

Raise the price 20% to \$12:

$$\% \Delta Q = 0.5 \times 20\% = 10\% \text{ fall} \Rightarrow Q = 100 - 10 = 90$$

$$TR_2 = 12 \times 90 = \$1,080$$

Same price move, opposite outcome. The only thing that changed was the elasticity.

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Slide 40

Working through it

Everything is identical except one number. Same 100 units, same \$10, same 20% price rise. Only the elasticity has changed, from 2 to 0.5. Watch what that alone does.

The quantity change. Same formula as before:

$$\% \Delta Q = |E_d| \times \% \Delta P = 0.5 \times 20\% = 10\%$$

A 10% fall from 100 units removes 10 units, leaving 90.

The new revenue. $12 \times 90 = \$1,080$, so revenue *rose* by \$80.

The same decomposition as before. The firm gained \$2 on each of 90 units, which is +\$180. It lost 10 units earning \$10 each, which is −\$100. Net: $180 - 100 = +\$80$. Both effects are still present and still fighting; this time the first is larger.

Why this pair of frames matters. Put them side by side. Same firm, same decision, same percentage price rise — and opposite outcomes. The only difference in the entire calculation was the elasticity. That is the argument for knowing it before you move your prices, not after.

Plug the numbers in — revenue along the coffee curve

Plug the numbers in — revenue along the coffee curve				
P (\$)	Q (000s)	$TR = P \times Q$ (\$000s)	$ E_d $	Region
1	60	$1 \times 60 = 60.0$	0.17	inelastic
2	50	$2 \times 50 = 100.0$	0.40	inelastic
3	40	$3 \times 40 = 120.0$	0.75	inelastic
3.50	35	$3.5 \times 35 = 122.5$	1.00	unit elastic
4	30	$4 \times 30 = 120.0$	1.33	elastic
5	20	$5 \times 20 = 100.0$	2.50	elastic
6	10	$6 \times 10 = 60.0$	6.00	elastic

Climbing through the inelastic rows, TR rises. Past \$3.50, into the elastic rows, it falls. The peak sits exactly at $|E_d| = 1$.

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Slide 41

Working through it

This table has five columns and the slide explains none of them. Take one row — the $P = 4$ row — and build it from scratch.

Column 2, quantity. From the coffee demand equation $Q^d = 70 - 10P$. Substituting: $Q = 70 - 10(4) = 70 - 40 = 30$.

Column 3, total revenue. Definition: $TR = P \times Q = 4 \times 30 = 120$, in thousands of dollars.

Column 4, elasticity — the one the slide simply asserts. We are using the slope method from Part 2:

$$|E_d| = |\text{slope}| \times \frac{P}{Q}$$

The slope is the 10 sitting in front of P in the demand equation: each extra dollar costs 10

thousand cups. So

$$|E_d| = 10 \times \frac{4}{30} = \frac{40}{30} = 1.33$$

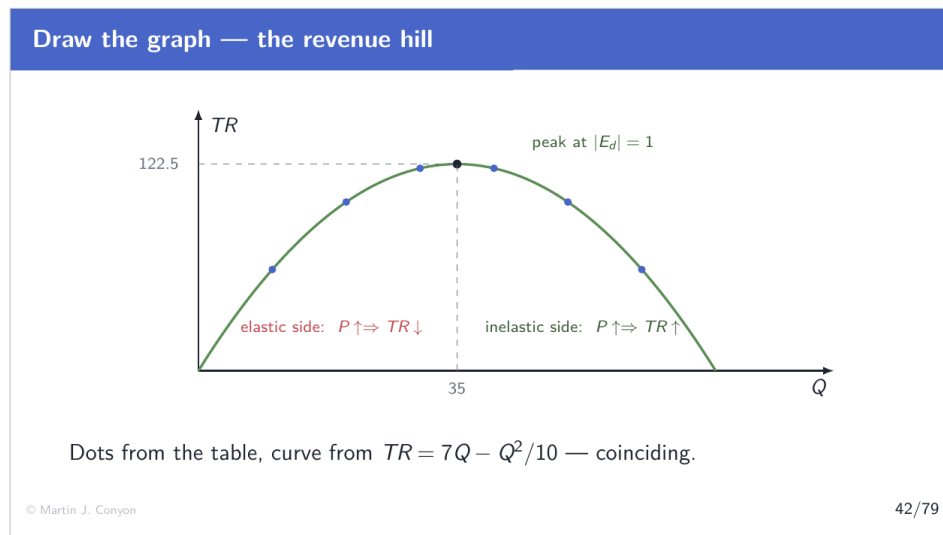
Every entry in that column is the same arithmetic. At $P = 1$: $10 \times \frac{1}{60} = 0.17$. At $P = 6$: $10 \times \frac{6}{10} = 6.00$. **The slope never changes — it is 10 in every row. What changes is the ratio P/Q** , which climbs steeply as you move up the curve, because P is rising while Q is falling. That is why elasticity is small at the bottom and enormous at the top of a straight-line demand curve.

Column 5, region. Purely a label for column 4: below 1 is inelastic, above 1 is elastic, exactly 1 is the boundary.

Now read down the revenue column. 60, 100, 120, 122.5, then 120, 100, 60. It climbs, turns, and falls. And the turn is at the row where $|E_d| = 1$.

Why \$3.50 and not some other price? Set the elasticity formula equal to one and solve. We need $10 \times P/Q = 1$, that is $Q = 10P$. Substitute the demand equation: $70 - 10P = 10P$, so $70 = 20P$ and $P = 3.50$. Then $Q = 70 - 35 = 35$. The unit-elastic point is not chosen or approximated — it is calculated, and it lands exactly halfway up the demand curve.

Draw the graph — the revenue hill



Slide 42

Working through it

What is on the axes. This figure is not a demand curve and it is easy to misread as one. The horizontal axis is quantity, as usual, but the vertical axis is **total revenue**, not price.

Where the curve's equation comes from. We want revenue written as a function of quantity alone. Start with the demand equation and make P the subject:

$$Q = 70 - 10P \implies 10P = 70 - Q \implies P = 7 - \frac{1}{10}Q$$

Then substitute into the definition of revenue:

$$TR = P \times Q = \left(7 - \frac{1}{10}Q\right)Q = 7Q - \frac{Q^2}{10}$$

That is the equation printed under the figure. It is not a new assumption — it is the definition of revenue with the demand curve substituted in.

Check it against the table. At $Q = 30$: $7(30) - 900/10 = 210 - 90 = 120$. That is the $P = 4$ row. At $Q = 35$: $245 - 122.5 = 122.5$, the peak.

Why the shape is a hill and not a line. The Q^2 term is what bends it. For small Q the first term dominates and revenue climbs; as Q grows the squared term catches up and eventually overwhelms it.

What the two labelled sides mean. Left of the peak, quantity is low and the price is high — and recall from the table that elasticity is *large* up there: $|E_d| = 6.00$ at $Q = 10$. So the left side is the elastic side, and raising the price moves you further left, down the near slope. Right of the peak, quantity is high and the price is low, and elasticity is below one: $|E_d| = 0.17$ at $Q = 60$. That is the inelastic side, and raising the price moves you left, up toward the summit. On this diagram a price rise always moves you leftward, because a higher price always means a smaller quantity.

Why the dots matter. The seven dots are the seven rows of the previous table, plotted. The curve is the equation. They coincide because they are the same object, computed two ways — which is the check.

The rule to carry

The rule to carry

Inelastic demand: raise the price, revenue rises.
Elastic demand: raise the price, revenue falls.
Revenue peaks where $|E_d| = 1$.

This one rule runs every revenue argument in economics — farm incomes, airline pricing, the choices facing any seller with market power.

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Slide 43

Working through it

Three lines, and they are the payoff of the whole part. The rectangle, the two effects, the seven-row table and the hill were all built to establish them.

Line 1, inelastic demand. $|E_d| < 1$ means quantity moves proportionately *less* than price. Raise the price and the price effect outweighs the quantity effect, so revenue rises.

Line 2, elastic demand. $|E_d| > 1$ means quantity moves proportionately *more*. The quantity effect wins, so the same price rise lowers revenue.

Line 3, and an answer owed since Part 2. Between the two regions sits $|E_d| = 1$, where the effects exactly cancel and revenue stops changing — the top of the hill. That is why *one* deserved to be the boundary of the vocabulary. It is not an arbitrary dividing line; it is the peak.

Where the rule turns up again. Any argument about whether raising a price raises money — farm incomes, airline fares, a seller with market power — is this rule wearing different labels.

Extensions — the same rule by calculus

Extensions — the same rule by calculus

Extension material — not needed for anything else here. The result above was already established numerically; this is the same result in one line, for those who have met derivatives.

Differentiate $TR = P \times Q$ along the demand curve:

$$\frac{dTR}{dP} = P \frac{dQ}{dP} + Q = -|E_d| Q + Q = Q(1 - |E_d|)$$

Positive when $|E_d| < 1$, zero at $|E_d| = 1$, negative when $|E_d| > 1$ — **the table, the hill, and the rule, in one line.**

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Slide 44

Working through it

Read this only if you have met derivatives. Nothing else depends on it. The revenue rule was already established twice, once in the table and once in the hill; this is the same result compressed into one line.

The differentiation. Revenue is $TR = P \times Q$, and along a demand curve Q depends on P , so the product rule gives

$$\frac{dTR}{dP} = P \frac{dQ}{dP} + Q$$

The first term is the quantity effect and the second is the price effect — the two forces named earlier, now written down.

Substituting the elasticity. Since $|E_d| = -\frac{dQ}{dP} \cdot \frac{P}{Q}$, we have $P \frac{dQ}{dP} = -|E_d| Q$, and therefore

$$\frac{dTR}{dP} = -|E_d| Q + Q = Q(1 - |E_d|)$$

Reading the result. Q is positive, so the sign is decided entirely by $1 - |E_d|$: positive below one, zero at one, negative above one. That is the three-line rule, the seven-row table and the shape of the hill, all in a single expression.

6. Part 6: The Elasticity of Supply

The seller's dial

The seller's dial

Price elasticity of supply (E_s). The percentage change in quantity supplied divided by the percentage change in price:

$$E_s = \frac{\% \Delta Q^s}{\% \Delta P}$$

Supply slopes **up**, so E_s is **positive** — no absolute value needed. The vocabulary transfers whole: elastic (> 1), unit ($= 1$), inelastic (< 1), and the two perfect extremes.

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Slide 48

Working through it

One word swapped. The price elasticity of supply is the percentage change in quantity *supplied* over the percentage change in price. Set it beside the demand definition and the only difference is which side of the market is responding.

Why there is no absolute value here. Supply curves slope upward, so a price rise brings a quantity rise. Both percentage changes carry the same sign and the ratio is positive on its own. There is nothing to strip.

What transfers unchanged. The whole vocabulary. Above one is elastic, quantity moving proportionately more than price. Exactly one is unit elastic. Below one is inelastic. And the same two extremes sit at the ends: $E_s = 0$ for a quantity that cannot respond at all, $E_s \rightarrow \infty$ for one that responds without limit.

What does not transfer. The *reasons*. Inelastic demand is about buyers having nowhere else to go; inelastic supply is about sellers being unable to produce more. The arithmetic is shared; the economics is not.

Computing it — same midpoint method

Computing it — same midpoint method

A bakery's sourdough supply: at $P = \$3$ a loaf, 3,000 loaves/week; at $P = \$4$, 5,000.

$$\% \Delta Q = \frac{2,000}{4,000} = 50\% \quad \% \Delta P = \frac{1}{3.50} = 28.6\%$$

$$E_s = \frac{50\%}{28.6\%} = 1.75 \text{ — elastic.}$$

Ovens run longer, an extra baker comes in: **quantity can chase the price.**

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Working through it

Which method, and why. We have two observed price–quantity pairs and no equation, so this is the midpoint method — exactly as on the demand side. Nothing new is being introduced.

The numerator. Quantity rises from 3,000 to 5,000, a change of 2,000. The average of the two is $(3,000 + 5,000)/2 = 4,000$. So $\% \Delta Q = 2,000/4,000 = 50\%$.

The denominator. Price rises from \$3 to \$4, a change of \$1. The average is $(3+4)/2 = 3.50$. So $\% \Delta P = 1/3.50 = 28.6\%$.

The ratio. $50/28.6 = 1.75$.

Why there is no absolute value here. On the demand side the two percentage changes always had opposite signs, so the raw ratio was negative and we took its absolute value by convention. Supply slopes upward, so both changes are positive and the ratio is positive on its own. Nothing to strip.

Reading the number. 1.75 exceeds one, so supply is elastic: quantity moves proportionately more than price. The economics is visible in the bakery itself — ovens can run longer and an extra baker can be called in, so output is free to chase the price.

What lets quantity chase the price

What lets quantity chase the price

1. Production flexibility. A print shop can add a shift within a day. A vineyard cannot add a harvest — the vines are planted.

2. Input availability. Expansion needs workers, materials, machines — **at hand**. If skilled labor is scarce, willing firms still cannot grow.

The hard limit: some quantities are **fixed by nature**. Beachfront land — no price mints more coastline. $E_s = 0$.

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Slide 50

Working through it

The question. Supply elasticity measures whether sellers can actually produce more when the price rises, so every determinant here is about production rather than preferences.

Determinant 1: production flexibility. A print shop can add a shift within a day, so its output follows the price almost at once. A vineyard cannot add a harvest this year — the vines were planted long ago and the season is the season — so its supply is very inelastic in the short run.

Determinant 2: input availability. Expanding needs workers, materials and machines to be *available*. A firm that wants to grow and cannot hire the skilled labor it needs will not grow, whatever the price. Willingness is not the constraint; the inputs are.

The hard limit. Some quantities are fixed by nature rather than by cost. No price mints more coastline, so the physical supply of beachfront land is perfectly inelastic, $E_s = 0$. This is the vertical panel of the figure that follows, and it is the one constraint that no amount of time relaxes.

Time, again

Time, again

3. Time horizon. The strongest determinant, exactly as on the demand side:

- **Short run** — capacity is fixed; output can only stretch within it. Supply **less** elastic.
- **Long run** — firms build plants, and **new firms enter**. Supply **more** elastic.

Almost every supply curve is more elastic in the long run.

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Slide 51

Working through it

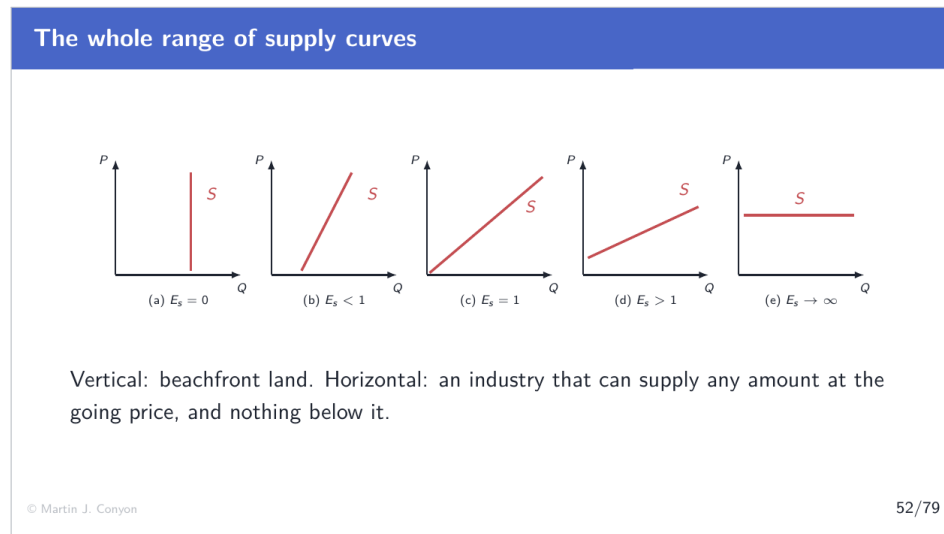
The same determinant as on the demand side, and again the strongest. What time changes is how much of the production process can be rearranged.

Short run. Capacity is whatever was built. Output can be stretched within the existing plant — longer hours, an extra shift — and no further, so supply is less elastic.

Long run. Two things become possible that were not before. Existing firms can build new capacity, and entirely new firms can enter the industry. The second is easy to overlook and is often the larger effect. Supply is more elastic.

The symmetry worth noticing. Both sides of the market obey the same law, and it is a human one rather than a technical one: give people time and they respond more. Buyers need time to find substitutes; sellers need time to build.

The whole range of supply curves



Slide 52

Working through it

The same five-panel layout as the demand version, in supply's color and sloping the other way. Read it left to right, from a quantity that cannot respond at all to one that responds without limit.

Panel (a), vertical. Quantity is fixed whatever the price: $E_s = 0$. Beachfront land is the standard case — no price mints more coastline.

Panel (c), the unit-elastic case. Note that this one passes through the origin. That is not decoration: any straight supply curve through the origin has $E_s = 1$ at every point, because price and quantity are then in fixed proportion, so a 1% rise in one is always a 1% rise in the other.

Panel (e), horizontal. Sellers will supply any quantity at this price and none at all below it: $E_s \rightarrow \infty$.

The one contrast worth carrying. On the demand side the vertical case was about buyers with no alternatives. Here the vertical case is about a quantity that physically cannot be increased. Different reason, same picture — which is worth noticing, because it means the diagram alone never tells you *why* something is inelastic.

7. Part 7: Tax Incidence: Who Really Pays

The check-writer is not the payer

The check-writer is not the payer

A city puts a **\$1 tax on every pizza sold, collected from the seller.**

Sellers will try to pass it on as a higher price. **Can they? How much of it?**

The law decides who mails the check. Elasticity decides who bears the burden.

Plan: **one market, one \$1 tax, two elasticity configurations.**

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Slide 56

Working through it

The setup. A \$1 tax on every pizza, and the statute says the seller hands it over. Taxes arranged this way are routinely described as taxes on business.

Why that description is not yet an answer. Sellers will try to recover the dollar through a higher price. Whether they can, and how much of it they recover, depends on whether buyers keep buying at the higher price — which is exactly what elasticity measures.

Two questions that must be kept apart. **Statutory incidence** is who is legally required to remit the tax, and that is written in the law. **Economic incidence** is whose pocket the money actually leaves, and that is settled by behavior. They need not be the same people, and the frames that follow show that they usually are not.

The plan. One market, one \$1 tax, and two configurations differing only in how responsive each side is. Holding the starting equilibrium and the tax fixed is what makes the comparison mean anything.

Configuration A — write the equations, find equilibrium**Configuration A — write the equations, find equilibrium**

Buyers hooked, sellers flexible:

$$Q^d = 100 - 5P \qquad Q^s = 20P - 50$$

$$100 - 5P = 20P - 50 \Rightarrow 150 = 25P \Rightarrow P^* = \$6, \quad Q^* = 100 - 5(6) = 70$$

Elasticities at the equilibrium (slope method):

$$|E_d| = 5 \times \frac{6}{70} = \mathbf{0.43} \qquad E_s = 20 \times \frac{6}{70} = \mathbf{1.71}$$

Demand inelastic, supply elastic. Prediction before solving: who bears more?

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Slide 57

Working through it

Why two equations, and where they come from. These are invented for the example, not derived from anything earlier. What matters is the shape they encode: demand with a small coefficient on P (5) is a curve buyers barely move along; supply with a large one (20) is a curve sellers move along freely. That is the configuration named at the top — buyers hooked, sellers flexible.

Finding the equilibrium. Set quantity demanded equal to quantity supplied and solve for P :

$$100 - 5P = 20P - 50 \implies 100 + 50 = 20P + 5P \implies 150 = 25P \implies P^* = 6$$

Then substitute back to get the quantity: $Q = 100 - 5(6) = 70$. Check it in the other equation too: $20(6) - 50 = 70$. Both agree, so the arithmetic is sound.

The elasticities. Slope method, exactly as in Part 2. The slope is the coefficient on P in each equation.

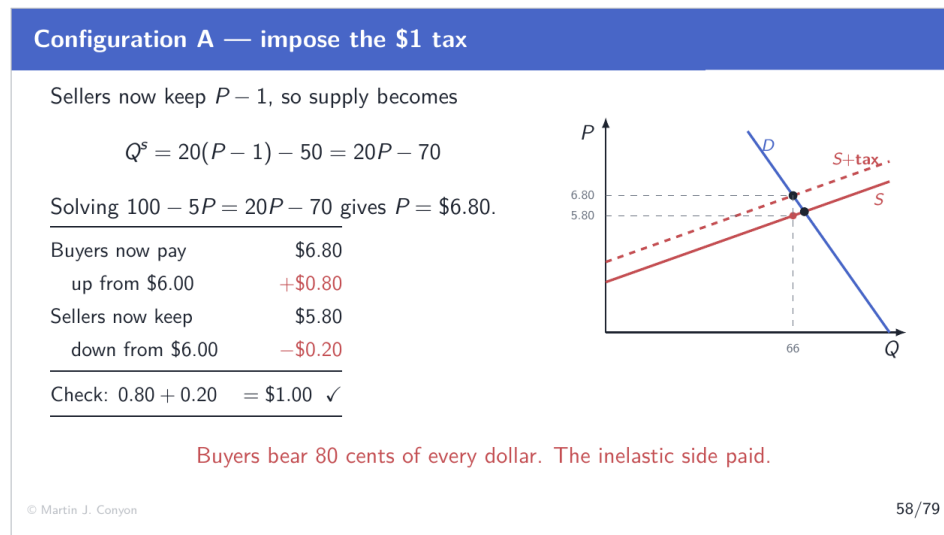
$$|E_d| = 5 \times \frac{6}{70} = \frac{30}{70} = 0.43 \qquad E_s = 20 \times \frac{6}{70} = \frac{120}{70} = 1.71$$

Demand is inelastic, supply is elastic — which is the whole point of choosing these particular

coefficients.

What to predict before reading on. If elasticity measures the ability to escape, the side that cannot escape should end up carrying the tax. Commit to that prediction now; the next frame tests it.

Configuration A — impose the \$1 tax



Slide 58

Working through it

Where the new supply equation comes from. This is the step the slide performs in one line and it is the crux of the whole calculation.

Sellers decide how much to supply based on what they *keep*, not what the buyer pays. With a \$1 tax, a buyer price of P leaves the seller with $P - 1$. So wherever the old supply equation had P , it now has $P - 1$:

$$Q^S = 20(P - 1) - 50 = 20P - 20 - 50 = 20P - 70$$

Solving again. Demand is untouched, so set the two equal as before:

$$100 - 5P = 20P - 70 \implies 170 = 25P \implies P = 6.80$$

Reading the table. The buyers' price is the P we just solved for: \$6.80, up 80 cents from \$6.00. The sellers' take is that price minus the tax: $6.80 - 1.00 = \$5.80$, down 20 cents.

Why the check matters. The two burdens must sum to exactly the tax: $0.80 + 0.20 = 1.00$. They must, because the buyer's price minus the seller's take is the tax by definition. If your two burdens do not sum to the tax, your algebra is wrong and you have just found out for free.

What the figure shows. The dashed red line is the new supply curve, sitting exactly \$1 vertically above the old one — the tax is a vertical shift, because it adds a fixed amount to what sellers must receive at every quantity. The equilibrium slides up the unchanged demand curve, from 70 to 66.

The verdict. The law collected the tax from sellers. Buyers paid 80% of it. The check-writer and the payer are different people.

Configuration B — swap the elasticities, repeat everything

Configuration B — swap the elasticities, repeat everything

Same equilibrium, roles reversed — buyers flexible, sellers stuck:

$$Q^d = 190 - 20P \quad Q^s = 5P + 40 \quad \Rightarrow \quad P^* = \$6, Q^* = 70$$

$$|E_d| = 20 \times \frac{6}{70} = \mathbf{1.71} \quad E_s = 5 \times \frac{6}{70} = \mathbf{0.43}$$

Same \$1 tax: $Q^s = 5(P - 1) + 40 \Rightarrow 190 - 20P = 5P + 35 \Rightarrow P = \6.20

Buyers pay \$6.20	+\$0.20
Sellers keep \$5.20	-\$0.80
Check: $0.20 + 0.80 = \$1.00$	✓

Same equilibrium, same tax — the burden flipped.

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Slide 59

Working through it

What is being held constant, and what is being swapped. This is a controlled experiment, so read it as one. The equilibrium is deliberately the same — \$6 and 70 — so that nothing about the starting point can explain the difference. Only the two coefficients have traded

places: demand now carries the 20 and supply the 5.

Confirming the equilibrium is genuinely unchanged.

$$190 - 20P = 5P + 40 \implies 150 = 25P \implies P = 6, \quad Q = 190 - 120 = 70$$

Identical to Configuration A.

The elasticities have swapped exactly.

$$|E_d| = 20 \times \frac{6}{70} = 1.71 \quad E_s = 5 \times \frac{6}{70} = 0.43$$

The two numbers from Configuration A, exchanged.

The tax, same procedure. Sellers keep $P - 1$, so $Q^s = 5(P - 1) + 40 = 5P + 35$. Setting demand equal to it:

$$190 - 20P = 5P + 35 \implies 155 = 25P \implies P = 6.20$$

Buyers pay \$6.20, up only 20 cents. Sellers keep $6.20 - 1.00 = \$5.20$, down 80 cents. Check: $0.20 + 0.80 = 1.00$.

Why this frame settles the argument. Same equilibrium, same tax, same law, same procedure. The only thing that differs between the two configurations is the relative responsiveness of buyers and sellers — and the burden reversed with it.

The incidence rule

The incidence rule				
The more inelastic side of the market bears more of the tax.				
	$ E_d $	E_s	Buyers bear	Sellers bear
Config. A	0.43	1.71	\$0.80	\$0.20
Config. B	1.71	0.43	\$0.20	\$0.80
Inelastic means nowhere to run : buyers who won't cut back, or sellers who can't cut production, absorb the price move. Who mails the check never mattered.				
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Working through it

The table is the experiment's result, in four numbers. Read it as a before-and-after of the swap: the elasticity columns exchange, and so do the burden columns. Nothing else in the two configurations differed.

Why inelasticity attracts the burden. Elasticity measures the ability to walk away. An elastic side has somewhere else to go — buyers to another good, sellers to another product line — so the price cannot move far against them before they simply leave. An inelastic side has nowhere to go, so the price can move against it a long way before behavior changes. The tax settles on whoever cannot escape it.

A sharper version of the rule. The burdens divide in the *inverse* ratio of the elasticities. In Configuration A,

$$\frac{E_s}{|E_d|} = \frac{120/70}{30/70} = 4$$

and the burden split is \$0.80 to \$0.20 — also 4 to 1. (The rounded values, 1.71/0.43, give 3.98; the exact fractions give 4.) This holds in the Active Learning exercise too, and it is worth checking there yourself.

The policy consequence. A legislature can decide who writes the check. It cannot decide who bears the cost. That is settled by the elasticities, which are facts about behavior and entirely outside the legislature's control — which is why “a tax on business” is a claim about paperwork rather than about economics.

Half a story — deliberately

Half a story — deliberately

The **who**: relative elasticities split the burden.

The other half, which belongs with the study of government policy:

- the **tax wedge** on the diagram,
- **revenue** the government collects,
- the trades the tax **prevents** — note Q fell $70 \rightarrow 66$ in *both* configurations.

Neither half repeats the other. Together they are the tax analysis.

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Working through it

What has been established. The *who*. With a per-unit tax in a competitive market, the relative elasticities split the burden, and the less elastic side bears more of it.

What has not. Three further pieces belong with the study of government policy: the tax drawn as a wedge between the price buyers pay and the price sellers keep; the revenue the government collects, which is the tax times the quantity still traded; and the trades the tax prevents.

The number on the slide that points at the missing half. Quantity fell from 70 to 66 in *both* configurations, whichever side bore the burden. Those four units are pizzas a buyer valued above what a seller would have accepted, and which now are not made. No part of the burden calculation counted them, because that calculation is about the transactions that still happen.

Why the two halves do not overlap. Who pays is a question about distribution; what is lost is a question about efficiency. A sensible verdict on any tax needs both.

8. Part 8: Applications

Good news arrives in the wheat market

Good news arrives in the wheat market

A better wheat variety raises yields. **Supply shifts right; demand — a staple, few substitutes — stays put and is inelastic.**

The market, before:

$$Q^d = 130 - 10P \quad Q^s = 50P - 50$$

$$130 - 10P = 50P - 50 \Rightarrow P^* = \$3, \quad Q^* = 100 \quad TR = \$300$$

Predict: bigger harvests — are farmers richer or poorer?

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Working through it

What the story sets up. Two things move in opposite directions here, and the frame is arranging them. A better variety means more wheat can be grown at any price, so *supply* shifts right. Demand for a staple with few substitutes does not move at all, and is steep.

Solving the starting equilibrium.

$$130 - 10P = 50P - 50 \Rightarrow 180 = 60P \Rightarrow P = 3$$

Substituting back: $Q = 130 - 10(3) = 100$. Check the other equation: $50(3) - 50 = 100$. Agreed.

Where the revenue figure comes from. $TR = P \times Q = 3 \times 100 = 300$. The units are dollars per bushel and millions of bushels, so this is \$300 million.

Why demand being inelastic is the loaded gun. Look at the coefficients. Demand loses only 10 units per dollar; supply gains 50. Demand is by far the steeper curve. From Part 5 you already know what a steep demand curve does to revenue when the price moves — and the supply shift is about to move it.

Commit to a prediction. Every farmer harvests more. Does the group earn more?

The same market, after

The same market, after

New supply: $Q^s = 50P + 10$. Demand unchanged.

$$130 - 10P = 50P + 10 \Rightarrow 120 = 60P \Rightarrow P^* = \$2, \quad Q^* = 110$$

	P	Q	$TR = P \times Q$
Before	\$3	100	\$300
After	\$2	110	\$220
Change	-33%	+10%	-\$80

More wheat, less revenue. Inelastic demand: the price fell three times as fast as quantity rose.

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Working through it

Where the new supply equation comes from. The better variety adds 60 to the quantity supplied at every price: -50 becomes $+10$. Demand is untouched.

Solving the new equilibrium.

$$130 - 10P = 50P + 10 \Rightarrow 120 = 60P \Rightarrow P = 2$$

and $Q = 130 - 10(2) = 110$. Check: $50(2) + 10 = 110$. Agreed.

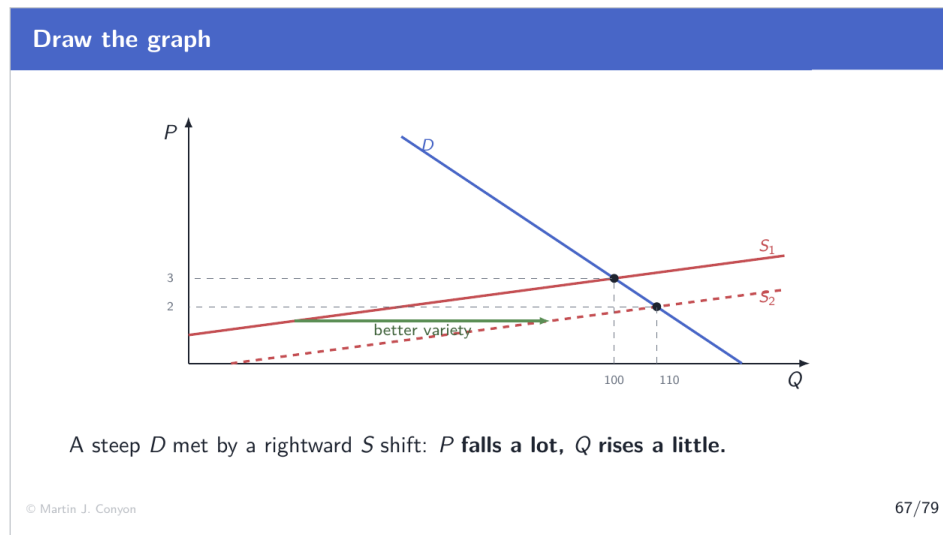
Building the table row by row. Revenue before: $3 \times 100 = 300$. Revenue after: $2 \times 110 = 220$. The change is $220 - 300 = -80$.

Where the percentages come from. Price fell from 3 to 2, so $-1/3 = -33\%$. Quantity rose from 100 to 110, so $+10/100 = +10\%$.

The sentence that explains everything. The price fell more than three times as fast, in percentage terms, as the quantity rose. Revenue is the product of the two, so it fell. And that ratio — price moving proportionately more than quantity — is precisely the definition of inelastic demand. The revenue rule from Part 5 is not being applied loosely here; it is the same arithmetic.

The uncomfortable conclusion. Every individual farmer must adopt the better variety, because refusing simply means growing less at the same low price. All of them adopting is what produces the low price. Each acts sensibly; the group ends up poorer.

Draw the graph



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Working through it

What is on the axes. Quantity horizontally, price vertically — an ordinary supply and demand diagram, with the supply curve drawn twice.

Which curve is which. The blue downward line is demand, unmoved throughout. S_1 is the original supply curve; the dashed S_2 is supply after the better variety, sitting to the *right* of it, because more is supplied at every price.

What to look at first. Whether demand is inelastic at the starting equilibrium — which is a calculation, not an impression:

$$|E_d| = 10 \times \frac{3}{100} = 0.30$$

comfortably below one. On this diagram that shows up as a steep demand curve, but steepness alone never settles it: the same straight line is elastic near its top, as Part 3 showed.

Reading the movement. The equilibrium slides from the upper dot down to the lower one. Notice the *shape* of that movement: a long way down and only a little to the right. Price collapsed; quantity crept.

Where you can see revenue in the picture. From Part 5, revenue is the rectangle under the equilibrium point. The first rectangle is 100×3 ; the second is 110×2 . It lost far more height than it gained width, so its area shrank. The steeper the demand curve, the more true that is — which is the geometric statement of the same rule.

This paradox emptied the countryside

This paradox emptied the countryside

In 1900, about **40% of the American labor force farmed**. By 2020, **1.4%**.

Two centuries of better seed, machinery, fertilizer — supply marching right against **inelastic food demand**. Prices fell, farm revenues fell, and labor moved to town.

Consumers captured the gains from farm technology. Farmers, as a group, did not.

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Working through it

The claim. The wheat calculation covered one market over one growing season. This frame scales the same mechanism up to two centuries.

The mechanism. Better seed, machinery and fertilizer each allow more food to be grown at any given price, so supply moves steadily rightward. Demand for food is inelastic: there is little substitute for eating, and people do not eat proportionately more as food gets cheaper. A rightward supply shift against inelastic demand pushes the price down proportionately more than it pushes quantity up, so revenue falls — exactly the wheat arithmetic, repeated for two hundred years.

Where the labor went. Falling revenue, alongside machinery that replaced hands, meant farming supported far fewer people. About 40% of the American labor force farmed in 1900; by 2020 direct on-farm employment was around 1.4% of U.S. employment.

Who gained. The technology worked and its gains were real — they went to consumers, as cheap food. Each individual farmer still had to adopt every improvement, because refusing meant growing less at the same low price. All of them adopting is what produced the low price.

The same logic, darker: the war on drugs

The same logic, darker: the war on drugs

Interdiction — seize supply at the border. S shifts left. Addiction makes demand **inelastic**: price soars, quantity barely falls, so **total spending** $P \times Q$ **rises** — and much of it is financed by theft. **More drug-related crime, not less.**

Education — shift *demand* left instead. Price falls *and* quantity falls: **spending falls unambiguously.**

Same goal, opposite predictions — separated only by which curve the policy moves.

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Working through it

Why this belongs here. It is the revenue rule from Part 5, applied with total *spending by buyers* in place of revenue to sellers. They are the same rectangle, $P \times Q$.

Interdiction, which moves supply. Seizing drugs at the border shifts supply left, so price rises and quantity falls. Addiction makes demand inelastic, so price rises proportionately more than quantity falls and total spending goes *up*. Where that spending is financed by theft, higher spending means more property crime — so the policy can increase the harm it was aimed at.

Education, which moves demand. A policy that reduces the desire to buy shifts demand left

instead. Price *and* quantity now fall together, so their product falls whatever the elasticity happens to be. There is no ambiguity left to resolve.

The transferable lesson. Two policies, one goal, opposite predictions. The difference is not intentions. It is which curve the policy moves, and what the elasticities are. Ask those two questions before predicting the effect of any intervention.

9. Part 9: Income and Cross-Price Elasticity

Same dial, new knob: income

Same dial, new knob: income

Income elasticity of demand (E_I). The percentage change in quantity demanded divided by the percentage change in income:

$$E_I = \frac{\% \Delta Q^d}{\% \Delta I}$$

The sign carries the information — so here, keep it. Incomes rise 10%; store-brand instant noodles fall 3%:

$$E_I = \frac{-3\%}{+10\%} = -0.3$$

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Working through it

What has changed, and what has not. The machine is identical: a ratio of two percentage changes. Only the denominator has been swapped, from the price of the good to the buyer's income.

Where the worked number comes from. Incomes rise 10%, and quantity of store-brand noodles falls 3%. So

$$E_I = \frac{-3\%}{+10\%} = -0.3$$

Why the sign is kept this time. On the demand side we discarded the sign, because the law

of demand made it negative every single time — a sign that is always the same carries no information. Here it is not always the same. Some goods rise with income and some fall, so the sign distinguishes two genuinely different kinds of good. Discarding it would throw away the most useful part of the answer.

What this particular number says. People bought *fewer* store-brand noodles as they got richer. They did not stop eating; they moved to something they preferred. The next frame gives that behavior its name.

What the sign and size say

What the sign and size say		
E_I	Term	Example
$E_I < 0$	Inferior	store-brand noodles, bus rides
$0 < E_I < 1$	Necessity	groceries as a whole
$E_I > 1$	Luxury	foreign vacations

Inferior goods are **fled** as income rises; necessities grow **slower** than income; luxuries grow **faster**.

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Working through it

Two pieces of information in one number. The sign says which way quantity moved with income. The size says how fast.

The sign, and the first split. Negative means the good is **inferior**: buyers take less of it as they grow richer, having moved to something they prefer. Store-brand noodles and bus rides are the standard cases. Positive means **normal**: more income, more of it. Most goods are normal.

The size, and the second split, inside the normal goods. Between zero and one, quantity rises with income but more slowly than income does — a **necessity**. Above one, it rises

faster than income — a **luxury**. Groceries as a whole sit below one; foreign vacations sit above it.

A caution about the examples. These labels describe a particular group of buyers over a particular range of incomes. A bus ride is inferior to someone who can afford a car and not to someone who cannot. What is being classified is the behavior, not the object.

Same dial again: the other good's price

Same dial again: the other good's price

Cross-price elasticity (E_{XY}). The percentage change in the quantity of good X demanded, divided by the percentage change in the *price of good Y* :

$$E_{XY} = \frac{\% \Delta Q_X^d}{\% \Delta P_Y}$$

$E_{XY} > 0$	Substitutes	tea ↑ when coffee price rises
$E_{XY} < 0$	Complements	ink ↓ when printer price rises

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Working through it

The last knob, and the only one that leaves the good's own market. The cross-price elasticity asks how the quantity of good X responds when the price of a *different* good Y moves.

Which good goes where. The good whose *quantity* moved goes in the numerator; the good whose *price* moved goes in the denominator. Reversing them asks a different question, and the two answers are generally different numbers.

Positive means substitutes. Coffee gets dearer, buyers jump ship, tea sales rise. The two changes share a sign, so the ratio is positive. Two goods are substitutes when one can stand in for the other.

Negative means complements. Printers get dearer, fewer are bought, and ink sales fall with them. The two changes have opposite signs, so the ratio is negative. Two goods are

complements when they are consumed together, so discouraging one discourages the other.

Reading a cross-price number

Reading a cross-price number

Gasoline prices rise **20%**; the quantity of new cars demanded falls **5%**:

$$E_{XY} = \frac{-5\%}{+20\%} = -0.25$$

Negative $\Rightarrow$ **complements** — cars are useless without fuel. Small (0.25) $\Rightarrow$ the link is real but **loose**.

Sign gives the relationship; size gives its strength.

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Working through it

Reading the calculation. Two facts are given: the price of one good rose 20%, and the quantity of a *different* good fell 5%. Divide:

$$E_{XY} = \frac{-5\%}{+20\%} = -0.25$$

Which good goes where. The good whose *quantity* moved goes in the numerator; the good whose *price* moved goes in the denominator. Reversing them answers a different question, and the two answers are generally not the same number.

What the sign says. Negative. Gasoline got dearer and fewer cars were bought, so the two move in opposite directions — they are consumed together. That is the definition of complements, and it accords with common sense, since a car without fuel is an ornament.

What the size says. 0.25 is small. Car purchases do respond to gasoline prices, but weakly: a 20% jump in gasoline shifted car sales by only 5%. The link is real and it is loose.

The discipline to carry. Two questions, two parts of the answer. The sign names the

relationship; the magnitude measures its strength. Both come out of a single division, and neither is worth anything without the other.